1.1) REFERRED TO FORMULA SHEET. USED CALCULATOR FOR ARITHMETIC.

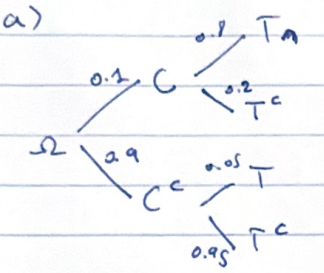
1.2) $P[A] = P[A|B_1]P[B_1] + P[A|B_2]P[B_2] + P[A|B_3]P[B_3]$
$= (0.2 \cdot 0.4) + (0.7 \cdot 0.1) + (0.5 \cdot 0.5) = 0.4$

$$P[B_3|A] = \frac{P[A|B_3] \cdot P[B_3]}{P[A]} = \frac{0.5 \cdot 0.5}{0.4} = 0.625$$

1.3) $P[C] = 0.1 \quad P[C^c] = 0.9 \quad T = \{$~~correct~~ positive test$\}$

$P[T^c|C] = 0.2 \quad P[T|C^c] = 0.05$
$P[T|C] = 0.8 \quad P[T^c|C^c] = 0.95$

a)



$$P[C|T^c] = \frac{P[C \cap T^c]}{P[T^c]} = \frac{0.1 \cdot 0.2}{(0.1 \cdot 0.2) + (0.9 \cdot 0.95)} = \frac{4}{175}$$

b) $$P[C|T] = \frac{P[C \cap T]}{P[T]} = \frac{0.1 \cdot 0.8}{(0.1 \cdot 0.8) + (0.9 \cdot 0.05)} = \frac{16}{25}$$

1.4) a) $P[B] = \frac{1}{2} \cdot \frac{1}{3}(1) + \cancel{2}\frac{1}{4} \cdot \frac{1}{\cancel{2}} + \cancel{\frac{1}{4} \cdot \frac{1}{2}} = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$

b) $P[G \cap C^c] = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{8} + \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{48} + \frac{3}{32} = \frac{5}{32}$

c) $$P[H_3|G \cap C^c] = \frac{P[H_3 \cap G \cap C^c]}{P[G \cap C^c]} = \frac{\frac{3}{\cancel{32}}}{\frac{5}{\cancel{32}}} = \frac{3}{5}$$

d) $$P[C|B] = \frac{P[C \cap B]}{P[B]} = \frac{\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2}}{\frac{5}{12}} = \frac{13}{20}$$

e) $$P[B|H_1 \cap C] = \frac{P[B \cap H_1 \cap C]}{P[H_1 \cap C]} = \frac{\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4}}{\frac{1}{2} \cdot \frac{1}{3}\left(\frac{3}{4} + \frac{7}{8} + \frac{5}{8}\right)} = \frac{1}{3}$$